SVKM's NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING / SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2021-22

Programme: B. Tech (Data Science) Year: II Semester: IV

Subject: Statistical Structures in Data and Inference

Marks: 100

Date: 07 June 2022

Time: 10.00 am to 1.00 pm
Durations: 3 (Hrs)
No. of Pages: 03

Re-Examination (2021-22)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) 1) Question No. _1____is compulsory.
2) 2) Out of remaining questions, attempt any __4____questions.
3) 3) **In all____5_ questions to be attempted.**
4) 4) All questions carry equal marks.
5) 5) **Answer to each new question to be started on a fresh page.**
6) 6) **Figures in brackets on the right hand side indicate full marks.**
7) 7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	marks
CO1; SO2; BL1	a.	What is Interval estimation? Define the following with an example: 1. Null Hypothesis 2. Alternate Hypothesis 3. Simple Hypothesis 4. Composite Hypothesis	5
CO3; SO1; BL2	b.	Define boot strapping and its use in statistics.	5
CO1; SO1; BL1	c.	Mention advantage and disadvantages of non-parametric test.	5

CO2; SO6; BL4	d.	Explain the concept of Autocorrelation.	5
Q2 CO3; SO1; BL5	a.	Define what is unbiasedness. Let X1,X2,X3 be a random sample of size 3 drawn from the population with mean Θ test whether the following estimators are unbiased:	10

		1. T1 = (X1+X2)/2 - X3 2. T2 = (X1 - X2 + X3)/3 3. T3 = (X1 - 3X2)/9 + 5X3 4. T4 = (X1 – X2 + X3)	
Q2 CO1; SO6; BL3	b.	1. Explain in brief what is Sufficiency, Efficiency 2. A random sample of size n is taken from uniform distribution over (0,θ). Prove that $2\bar{x}$ is an unbiased estimate for θ.	10
Q3 CO3; SO2; BL2	a.	1. What is MLE? Generalize the steps involved in getting the MLE of an estimate. 2. Mention Properties of MLE.	10
Q3 CO3; SO1; BL3	b.	1. Find MLE of parameter θ for the pdf f(x, θ) = (1- θ) θ^(x-1) 1) where x = 0,1,2…,∞ 2. A sample of 400 male students is found to have a mean height of 67.47 inches. Can it be reasonably regarded as a sample from a large population with mean height 69.39 inches and standard deviation 2.30 inches. Test for 5% level of significance.	10
Q4 CO1; SO6; BL6	a.	1. Explain all the assumptions of linear regression and mention and mention the model of linear regression with explaining all the variables. 2. Mention the properties of OLS estimators.	10
Q4 CO4; SO7; BL3	b.	1. Explain are the 4 possibilities to make the decision to accept or to reject the Null hypothesis? explain with the help of a table. 2. Explain the concept of autocorrelation.	10

Q5 CO1; SO6; BL2	a.	1. What is logistic distribution function? 2. What is its significance in logistic regression model? 3. Explain logit transformation of logistic regression model	10
Q5	b.	1. Explain Adjusted R-Square. 2. Explain Heteroscedasticity	10
CO4; SO7; BL3			
Q6 CO1; SO1; BL1	a.	For One simple sign test mention the: 1. Assumptions 2. Hypothesis to be tested 3. Test statistic 4. Decision Rule	10
Q6 CO1; SO1; BL2	b.	For Mann Whitney test mention the: 1. Assumptions 2. Hypothesis to be tested 3. Test statistic 4. Decision Rule	10